

Bibliography for Master's Project: PFAS Removal in Wastewater Treatment: Chain-Length, Membrane Performance, and Technology Selection with Fuzzy MCDA

Literature Cited on Presentation

Literature Cited on Report

- Ackerman Grunfeld, D., D. Gilbert, J. Hou, A. M. Jones, M. J. Lee, T. C. G. Kibbey, and D. M. O'Carroll. 2024. Underestimated burden of per- and polyfluoroalkyl substances in global surface waters and groundwaters. *Nature Geoscience* **17**: 340–346. DOI:10.1038/s41561-024-01402-8.
- Aedan, Y., A. Altaee, V. Mueller, H. K. Shon, L. Alsaka, and S. J. Zaidi. 2026. PFAS-contaminated groundwater treatment by forward osmosis using pressure-stimuli-responsive nanofiltration membrane: Effects of chain length and operating conditions. *Desalination* **617**: 119425. DOI:10.1016/j.desal.2025.119425.
- Ahmad, A., and K. Tian. 2024. Transport of per-polyfluoroalkyl substances (PFAS) in high-density polyethylene (HDPE) geomembrane liners. *E3S Web Conf.* **569**.
- Ahmad, N. N. R., W. L. Ang, C. P. Leo, A. W. Mohammad, and N. Hilal. 2021. Current advances in membrane technologies for saline wastewater treatment: A comprehensive review. *Desalination* **517**: 115170. DOI:10.1016/j.desal.2021.115170.
- Al Aani, S., T. N. Mustafa, and N. Hilal. 2020. Ultrafiltration membranes for wastewater and water process engineering: A comprehensive statistical review over the past decade. *Journal of Water Process Engineering* **35**. DOI:10.1016/j.jwpe.2020.101241.
- Al-Maqtari, Q. A., A. A. Mahdi, N. Othman, A. E. Noman, L. M. Alsubhi, W. AlAnsi, S. M. Asharuddin, M. k. Talib, S. Supramaniam, and A. Al-Gheethi. 2025. Comprehensive review of per- and polyfluoroalkyl substances in drinking water: Physiochemical properties, sources, health impacts, regulatory limits, and future perspectives. *Results in Engineering* **28**: 107169. DOI:10.1016/j.rineng.2025.107169.
- Bellona, C., D. Heil, C. Yu, P. Fu, and J. E. Drewes. 2012. The pros and cons of using nanofiltration in lieu of reverse osmosis for indirect potable reuse applications. *Separation and Purification Technology* **85**: 69–76. DOI:10.1016/j.seppur.2011.09.046.
- Bentel, M. J., Y. Yu, L. Xu, Z. Li, B. M. Wong, Y. Men, and J. Liu. 2019. Defluorination of Per- and Polyfluoroalkyl Substances (PFASs) with Hydrated Electrons: Structural Dependence and Implications to PFAS Remediation and Management. *Environmental Science and Technology* **53**: 3718–3728. DOI:10.1021/acs.est.8b06648.
- Björnsdotter, M. K., L. W. Y. Yeung, A. Kärrman, and I. Ericson Jogsten. 2020. Challenges in the analytical determination of ultra-short-chain perfluoroalkyl acids and implications for environmental and human health. *Analytical and bioanalytical chemistry* **412**: 4785–4796. DOI:10.1007/s00216-020-02692-8.

- Boone, J. S., C. Vigo, T. Boone, C. Byrne, J. Ferrario, R. Benson, J. Donohue, J. E. Simmons, D. W. Kolpin, E. T. Furlong, and S. T. Glassmeyer. 2019. Per- and polyfluoroalkyl substances in source and treated drinking waters of the United States. *Science of The Total Environment* **653**: 359–369. DOI:10.1016/j.scitotenv.2018.10.245.
- Brendel, S., É Fetter, C. Staude, L. Vierke, and A. Biegel-Engler. 2018. Short-chain perfluoroalkyl acids: environmental concerns and a regulatory strategy under REACH. *Environmental Sciences Europe* **30**: 9. DOI:10.1186/s12302-018-0134-4.
- Brennan, N. M., A. T. Evans, M. K. Fritz, S. A. Peak, and H. E. von Holst. 2021. Trends in the Regulation of Per- and Polyfluoroalkyl Substances (PFAS): A Scoping Review. *International Journal of Environmental Research and Public Health* **18**. DOI:10.3390/ijerph182010900.
- Buck, R. C., J. Franklin, U. Berger, J. M. Conder, I. T. Cousins, P. D. Voogt, A. A. Jensen, K. Kannan, S. A. Mabury, and S. P. J. van Leeuwen. 2011. Perfluoroalkyl and polyfluoroalkyl substances in the environment: Terminology, classification, and origins. *Integrated Environmental Assessment and Management* **7**: 513–541. DOI:10.1002/ieam.258.
- Cantoni, B., G. Bergna, E. Baldini, F. Malpei, and M. Antonelli. 2024. PFAS in textile wastewater: An integrated scenario analysis for interventions prioritization to reduce environmental risk. *Process Safety and Environmental Protection* **183**: 437–445. DOI:10.1016/j.psep.2024.01.005.
- Casella, G. S., M. M. Miranda, G. P. Drumond, G. O. R. e. Silva, F. Zouaoui, G. S. Arcanjo, M. C. S. Amaral, E. C. de Paula, and V. R. Moreira. 2026. Recent developments in membrane treatment technologies for PFAS removal from surface and groundwater. *Journal of Environmental Chemical Engineering* **14**: 120598. DOI:10.1016/j.jece.2025.120598.
- Chang, C., A. S. Young, A. Keil, C. E. Mullins, D. Liang, S. Zhao, D. P. Jones, X. Hu, D. I. Walker, and A. J. White. 2025. Novel and legacy per- and polyfluoroalkyl substances in humans: Long-term temporal variability and metabolic perturbations. *Environment international* **201**: 109590. DOI:10.1016/j.envint.2025.109590.
- Chaudhary, M., M. Sela-Adler, A. Ronen, and O. Nir. 2023a. Efficient PFOA removal from drinking water by a dual-functional mixed-matrix-composite nanofiltration membrane. *npj Clean Water* **6**: 77. DOI:10.1038/s41545-023-00286-2.
- Chaudhary, M., M. Sela-Adler, A. Ronen, and O. Nir. 2023b. Efficient PFOA removal from drinking water by a dual-functional mixed-matrix-composite nanofiltration membrane. *npj Clean Water* **6**: 77. DOI:10.1038/s41545-023-00286-2.
- Chen, Q., J. Wang, Z. Su, L. Tian, F. Huang, T. Liu, N. Graham, G. Li, and W. Yu. 2024. Per- and polyfluoroalkyl substances (PFAS) at low concentration improve coagulation efficiency but induce higher membrane fouling in drinking water treatment. *Environmental Pollution* **363**. DOI:10.1016/j.envpol.2024.125201.
- Chen, W., M. Liu, M. Ding, L. Zhang, and S. Dai. 2023. Advanced thin-film composite polyamide membrane for precise trace short-chain PFAS sieving: Solution, environment and fouling effects. *Process Safety and Environmental Protection* **169**: 493–503. DOI:10.1016/j.psep.2022.11.036.

- Chen, W., F. Yang, E. Hu, C. Yang, C. Sun, and M. Li. 2022. Occurrence, fate and risk assessment of per- and polyfluoroalkyl substances in wastewater treatment plants in Shaanxi, China. *Environmental Pollution* **314**: 120226. DOI:10.1016/j.envpol.2022.120226.
- Chen, X., A. Vanangamudi, J. Wang, J. Jegatheesan, V. Mishra, R. Sharma, S. R. Gray, J. Kujawa, W. Kujawski, F. Wicaksana, and L. F. Dumée. 2020. Direct contact membrane distillation for effective concentration of perfluoroalkyl substances – Impact of surface fouling and material stability. *Water research* **182**. DOI:10.1016/j.watres.2020.116010.
- Deringer, A. 2024. Organic Fouling Behavior In Nanofiltration Membranes For The Removal of Per- and Polyfluoroalkyl Substances. Dissertation/Thesis. The University of Wisconsin-Milwaukee. .
- Ding, L., Y. Wei, L. Li, T. Zhang, H. Wang, J. Xue, L. Ding, S. Wang, J. Caro, and Y. Gogotsi. 2018. MXene molecular sieving membranes for highly efficient gas separation. *Nature Communications* **9**: 155. DOI:10.1038/s41467-017-02529-6.
- Dischinger, S. M., D. J. Miller, D. A. Vermaas, and R. S. Kingsbury. 2024. Unifying the Conversation: Membrane Separation Performance in Energy, Water, and Industrial Applications. *ACS ES&T Engineering* **4**: 277–289. DOI:10.1021/acsestengg.3c00475.
- Fan, H., K. Xiao, S. Mu, Y. Zhou, J. Ma, X. Wang, and X. Huang. 2018. Impact of membrane pore morphology on multi-cycle fouling and cleaning of hydrophobic and hydrophilic membranes during MBR operation. *Journal of Membrane Science* **556**: 312–320. DOI:10.1016/j.memsci.2018.04.014.
- Fang, F., S. Chen, K. Shi, S. Xu, Z. Yi, L. Lei, L. Zhuang, H. Wan, and Z. Xu. 2024. Hydrophilic membranes for effective removal of PFAS from water: Anti-fouling, durability, and reusability. *Separation and Purification Technology* **348**. DOI:10.1016/j.seppur.2024.127379.
- Fang, F., S. Chen, K. Shi, S. Xu, Z. Yi, L. Lei, L. Zhuang, H. Wan, and Z. Xu. 2024. Hydrophilic membranes for effective removal of PFAS from water: Anti-fouling, durability, and reusability. *Separation and Purification Technology* **348**: 127379. DOI:10.1016/j.seppur.2024.127379.
- Fenton, S. E., A. Ducatman, A. Boobis, J. C. DeWitt, C. Lau, C. Ng, J. S. Smith, and S. M. Roberts. 2021. Per- and Polyfluoroalkyl Substance Toxicity and Human Health Review: Current State of Knowledge and Strategies for Informing Future Research. *Environmental toxicology and chemistry* **40**: 606–630. DOI:10.1002/etc.4890.
- Fujioka, T., H. Takeuchi, H. Tahara, H. Murakami, and S. Boivin. 2024. Effects of functional groups of polyfluoroalkyl substances on their removal by nanofiltration. *Water Research X* **24**: 100233. DOI:10.1016/j.wroa.2024.100233.
- Gagliano, E., M. Sgroi, P. P. Falciglia, F. G. A. Vagliasindi, and P. Roccaro. 2020. Removal of poly- and perfluoroalkyl substances (PFAS) from water by adsorption: Role of PFAS chain length, effect of organic matter and challenges in adsorbent regeneration. *Water research* **171**: 115381. DOI:10.1016/j.watres.2019.115381.
- Gobelius, L., L. Glimstedt, J. Olsson, K. Wiberg, and L. Ahrens. 2023. Mass flow of per- and polyfluoroalkyl substances (PFAS) in a Swedish municipal wastewater network and wastewater treatment plant. *Chemosphere* **336**: 139182. DOI:10.1016/j.chemosphere.2023.139182.

- Griffin, A. M., C. Bellona, and T. J. Strathmann. 2024. Rejection of PFAS and priority co-contaminants in semiconductor fabrication wastewater by nanofiltration membranes. *Water research* **262**: 122111. DOI:10.1016/j.watres.2024.122111.
- Hamza, M., R. T. Ayinla, I. Elsayed, and E. B. Hassan. 2025. Understanding PFAS Adsorption: How Molecular Structure Affects Sustainable Water Treatment. *Environments* **12**. DOI:10.3390/environments12090330.
- Hang, X., X. Chen, J. Luo, W. Cao, and Y. Wan. 2015. Removal and recovery of perfluorooctanoate from wastewater by nanofiltration. *Separation and Purification Technology* **145**: 120–129. DOI:10.1016/j.seppur.2015.03.013.
- Hasan, R., J. Chen, P. Mojahednia, S. H. Samaei, and J. Xue. 2025. Comparative Analysis of Commercial and Novel High-Pressure Membranes for Perfluoroalkyl and Polyfluoroalkyl Substances (PFAS) Removal. *Water environment research : a research publication of the Water Environment Federation* **97**: e70157. DOI:10.1002/wer.70157.
- Hosseinzadeh, A., J. L. Zhou, J. Zyaie, N. AlZainati, I. Ibrar, and A. Altaee. 2022. Machine learning-based modeling and analysis of PFOS removal from contaminated water by nanofiltration process. *Separation and Purification Technology* **289**: 120775. DOI:10.1016/j.seppur.2022.120775.
- Houtz, E. F., R. Sutton, J. Park, and M. Sedlak. 2016a. Poly- and perfluoroalkyl substances in wastewater: Significance of unknown precursors, manufacturing shifts, and likely AFFF impacts. *Water research* **95**: 142–149. DOI:10.1016/j.watres.2016.02.055.
- Houtz, E. F., R. Sutton, J. Park, and M. Sedlak. 2016b. Poly- and perfluoroalkyl substances in wastewater: Significance of unknown precursors, manufacturing shifts, and likely AFFF impacts. *Water research* **95**: 142–149. DOI:10.1016/j.watres.2016.02.055.
- Hua, Y., Y. Li, P. Li, B. Chai, Z. Zhou, and J. Wu. 2026. Enhanced removal of anionic Per- and polyfluoroalkyl substances (PFAS) by electrically assisted nanofiltration: Operational parameter optimization and mechanistic insights. *Journal of hazardous materials* **504**: 141395. DOI:10.1016/j.jhazmat.2026.141395.
- Hussain, H. N., M. I. Jilani, F. Imtiaz, T. Ahmed, M. B. Arshad, M. Mudassar, and M. N. Sharif. 2025. Advances in the removal of Polyfluoroalkyl Substances (PFAS) from water using destructive and non-destructive methods. *Green Analytical Chemistry* **12**: 100225. DOI:10.1016/j.greeac.2025.100225.
- Jin, T., M. Peydayesh, and R. Mezzenga. 2021. Membrane-based technologies for per- and polyfluoroalkyl substances (PFASs) removal from water: Removal mechanisms, applications, challenges and perspectives. *Environment international* **157**: 106876. DOI:10.1016/j.envint.2021.106876.
- Kandel, D. R., D. Kwak, S. Lee, Y. J. Lim, S. Subedi, and J. Lee. 2024. Harnessing natural antifouling agents for enhancing water and wastewater treatment membranes. *Separation and Purification Technology* : 130254.
- Kannan, U., M. K. Ravichandran, M. Monisha, S. Krishnasamy, Valliammai, P. Prabakar, D. Darshini, and G. Sunantha. 2026. Short-chain PFAS: A global challenge - Environmental fate, health risks,

- and promising removal technologies. *Results in Chemistry* **20**: 103000. DOI:10.1016/j.rechem.2025.103000.
- Karbassiyazdi, E., M. Kasula, S. Modak, J. Pala, M. Kalantari, A. Altaee, M. R. Esfahani, and A. Razmjou. 2023. A juxtaposed review on adsorptive removal of PFAS by metal-organic frameworks (MOFs) with carbon-based materials, ion exchange resins, and polymer adsorbents. *Chemosphere* **311**: 136933. DOI:10.1016/j.chemosphere.2022.136933.
- Kwiatkowski, C. F., D. Q. Andrews, L. S. Birnbaum, T. A. Bruton, J. C. DeWitt, D. R. U. Knappe, M. V. Maffini, M. F. Miller, K. E. Pelch, A. Reade, A. Soehl, X. Trier, M. Venier, C. C. Wagner, Z. Wang, and A. Blum. 2020. Scientific Basis for Managing PFAS as a Chemical Class. *Environmental Science & Technology Letters* **7**: 532–543. DOI:10.1021/acs.estlett.0c00255.
- Kwon, H. -, H. -. Kim, Y. -. Park, K. -. Seok, J. -. Oh, and S. -. Choi. 2017. Updated national emission of perfluoroalkyl substances (PFASs) from wastewater treatment plants in South Korea. *Environmental Pollution* **220**: 298–306. DOI:10.1016/j.envpol.2016.09.063.
- Lai, Y., T. Liu, and C. Hwang. 1994. TOPSIS for MODM. *European Journal of Operational Research* **76**: 486–500. DOI:10.1016/0377-2217(94)90282-8.
- Lanjewar, S., A. Mukherjee, P. Khandewal, A. K. Ghosh, A. Mullick, S. Moulik, and A. Roy. 2021. Thermodynamics of synthesis and separation performance of Interfacially polymerized “loose” reverse osmosis membrane: Benchmarking for greywater treatment. *Chemical Engineering Journal* **417**: 127929. DOI:10.1016/j.cej.2020.127929.
- Lee, T., T. F. Speth, and M. N. Nadagouda. 2022. High-pressure membrane filtration processes for separation of Per- and polyfluoroalkyl substances (PFAS). *Chemical Engineering Journal* **431**. DOI:10.1016/j.cej.2021.134023.
- Lee, Y., G. Moon, H. Cha, J. Lee, and E. E. Kwon. 2026. Perfluoroalkyl chain-length-length-dependent environmental fate and treatment outcomes of PFAS in water. *npj Clean Water* . DOI:10.1038/s41545-026-00568-5.
- Léniz-Pizarro, F., R. J. Vogler, P. Sandman, N. Harris, L. E. Ormsbee, C. Liu, and D. Bhattacharyya. 2022. Dual-Functional Nanofiltration and Adsorptive Membranes for PFAS and Organics Separation from Water. *ACS EST Water* **2**: 863–872. DOI:10.1021/acsestwater.2c00043.
- Leung, S. C. E., D. Wanninayake, D. Chen, N. Nguyen, and Q. Li. 2023. Physicochemical properties and interactions of perfluoroalkyl substances (PFAS) - Challenges and opportunities in sensing and remediation. *Science of The Total Environment* **905**: 166764. DOI:10.1016/j.scitotenv.2023.166764.
- Li, F., C. Deng, C. Du, B. Yang, and Q. Tian. 2015. Fouling mechanism and cleanability of ultrafiltration membranes modified with polydopamine-graft-PEG. *Water SA* **41**: 448–456. DOI:10.4314/wsa.v41i4.03.
- Li, F., J. Duan, S. Tian, H. Ji, Y. Zhu, Z. Wei, and D. Zhao. 2020. Short-chain per- and polyfluoroalkyl substances in aquatic systems: Occurrence, impacts and treatment. *Chemical Engineering Journal* **380**: 122506. DOI:10.1016/j.cej.2019.122506.

- Li, S., L. Duan, Q. Gao, and H. Zhang. 2025. Performance impacts and removal potential of per- and poly-fluoroalkyl substances in osmotic microbial fuel cell. *Separation and Purification Technology* **357**. DOI:10.1016/j.seppur.2024.130236.
- Lin, D., M. Méndez, K. Paterson, A. Wong, D. Yee, R. Sutton, E. Houtz, M. Cousins, and L. Fono. 2024. Residential Wastewater as a Major Source of Per- and Polyfluoroalkyl Substances to Municipal Wastewater. *ACS EST Water* **4**: 4847–4857. DOI:10.1021/acsestwater.4c00507.
- Liu, C., X. Zhao, A. F. Faria, K. Y. Deliz Quiñones, C. Zhang, Q. He, J. Ma, Y. Shen, and Y. Zhi. 2022. Evaluating the efficiency of nanofiltration and reverse osmosis membrane processes for the removal of per- and polyfluoroalkyl substances from water: A critical review. *Separation and Purification Technology* **302**: 122161. DOI:10.1016/j.seppur.2022.122161.
- Liu, J., and J. A. Charbonnet. 2025. A Critical Review of PFAS Analysis, Occurrence, and Fate in Wastewater Treatment Plants. *Environmental science & technology* **59**: 25492–25517. DOI:10.1021/acs.est.5c09949.
- Liu, S., J. Kim, X. Xin, Z. Chen, X. Tong, Y. Zhao, J. Crittenden, and Y. Chen. 2025. Polyelectrolyte membranes for selective nutrient recovery from PFAS-contaminated wastewater in forward osmosis. *Water research* **287**: 124444. DOI:10.1016/j.watres.2025.124444.
- Lohmann, R., and R. J. Letcher. 2023. The universe of fluorinated polymers and polymeric substances and potential environmental impacts and concerns. *Current opinion in green and sustainable chemistry* **41**: 10.1016/j.cogsc.2023.100795. Epub 2023 Feb 27. DOI:10.1016/j.cogsc.2023.100795.
- Ma, Q., J. Zhang, G. Zhu, N. Ahuja, B. Khusid, and W. Zhang. 2024. Effects of surfactants, ion valency and solution temperature on PFAS rejection in commercial reverse osmosis (RO) and nanofiltration (NF) processes. *Journal of Water Process Engineering* **66**: 106039. DOI:10.1016/j.jwpe.2024.106039.
- Mabaso, N. S. N., C. S. Tshangana, and A. A. Muleja. 2024. Efficient Removal of PFASs Using Photocatalysis, Membrane Separation and Photocatalytic Membrane Reactors. *Membranes* **14**. DOI:10.3390/membranes14100217.
- Mastropietro, T. F., R. Bruno, E. Pardo, and D. Armentano. 2021. Reverse osmosis and nanofiltration membranes for highly efficient PFASs removal: overview, challenges and future perspectives. *Dalton Transactions* **50**: 5398–5410. DOI:10.1039/d1dt00360g.
- Mendoza-Apodaca, Y., S. Walsh, A. Rosu, and M. B. Shiflett. 2026. Membrane Technology for Hydrofluorocarbon Refrigerant Separation. *Industrial & Engineering Chemistry Research* **65**: 6395–6415. DOI:10.1021/acs.iecr.5c04781.
- Metz, J., P. Zuo, B. Wang, M. S. Wong, and P. J. J. Alvarez. 2022. Perfluorooctanoic acid Degradation by UV/Chlorine. *Environmental Science & Technology Letters* **9**: 673–679. DOI:10.1021/acs.estlett.2c00452.
- Moneta, B. G., M. L. Feo, M. Torre, P. Tratzi, S. E. Aita, C. M. Montone, E. Taglioni, S. Mosca, C. Balducci, M. Cerasa, E. Guerriero, F. Petracchini, C. Cavaliere, A. Laganà, and V. Paolini. 2023. Occurrence of per- and polyfluorinated alkyl substances in wastewater treatment plants in Northern Italy. *Science of The Total Environment* **894**: 165089. DOI:10.1016/j.scitotenv.2023.165089.

- Nădăban, S., S. Dzitac, and I. Dzitac. 2016. Fuzzy TOPSIS: A General View. *Procedia Computer Science* **91**: 823–831. DOI:10.1016/j.procs.2016.07.088.
- Ng, C., I. T. Cousins, J. C. DeWitt, J. Glüge, G. Goldenman, D. Herzke, R. Lohmann, M. Miller, S. Patton, M. Scheringer, X. Trier, and Z. Wang. 2021. Addressing Urgent Questions for PFAS in the 21st Century. *Environmental science & technology* **55**: 12755–12765. DOI:10.1021/acs.est.1c03386.
- O'Connor, J., N. S. Bolan, M. Kumar, A. S. Nitai, M. B. Ahmed, S. S. Bolan, M. Vithanage, J. Rinklebe, R. Mukhopadhyay, P. Srivastava, B. Sarkar, A. Bhatnagar, H. Wang, K. H. M. Siddique, and M. B. Kirkham. 2022. Distribution, transformation and remediation of poly- and per-fluoroalkyl substances (PFAS) in wastewater sources. *Process Safety and Environmental Protection* **164**: 91–108. DOI:10.1016/j.psep.2022.06.002.
- Ochando-Pulido, J. M., A. Martínez-Férez, and M. Stoller. 2018. Analysis of the Flux Performance of Different RO/NF Membranes in the Treatment of Agroindustrial Wastewater by Means of the Boundary Flux Theory. *Membranes* **9**: 2. doi: 10.3390/membranes9010002. DOI:10.3390/membranes9010002.
- OECD/UNEP Global PFC Group. 2013. United nations environment programme: synthesis paper on per- and polyfluorinated chemicals (PFCs), environment, health and safety, environment directorate. OECD. IOMC inter-organization program :.
- O'Hagan, D. 2008a. Understanding organofluorine chemistry. An introduction to the C-F bond. *Chemical Society Reviews* **37**: 308–319. DOI:10.1039/b711844a.
- O'Hagan, D. 2008b. Understanding organofluorine chemistry. An introduction to the C–F bond. *Chemical Society Reviews* **37**: 308–319. DOI:10.1039/B711844A.
- Riegel, M., B. Haist-Gulde, and F. Sacher. 2023. Sorptive removal of short-chain perfluoroalkyl substances (PFAS) during drinking water treatment using activated carbon and anion exchanger. *Environmental Sciences Europe* **35**: 12. DOI:10.1186/s12302-023-00716-5.
- Safulko, A., T. Y. Cath, F. Li, B. Tajdini, M. Boyd, R. P. Huehmer, and C. Bellona. 2023. Rejection of perfluoroalkyl acids by nanofiltration and reverse osmosis in a high-recovery closed-circuit membrane filtration system. *Separation and Purification Technology* **326**: 124867. DOI:10.1016/j.seppur.2023.124867.
- Salawu, O. A., Z. Han, N. L. Senehi, S. L. Roback, and A. S. Adeleye. 2025. Determination of household wastewater PFAS composition and concentrations via neighborhood sampling. *Journal of hazardous materials* **498**: 139834. DOI:10.1016/j.jhazmat.2025.139834.
- Scheepers, D., B. Chatillon, K. Nijmeijer, and Z. Borneman. 2021. Asymmetric layer-by-layer polyelectrolyte nanofiltration membranes with tunable retention. *Journal of Polymer Science* **59**: 1293–1304. DOI:10.1002/pol.20210166.
- Shyur, H., and H. Shih. 2006. A hybrid MCDM model for strategic vendor selection. *Mathematical and Computer Modelling* **44**: 749–761. DOI:10.1016/j.mcm.2005.04.018.
- Sodhi, B., and P. T V. 2012. A simplified description of Fuzzy TOPSIS. *arXiv preprint arXiv:1205.5098* .

- Soriano, Á, D. Gorri, and A. Urtiaga. 2019. Selection of High Flux Membrane for the Effective Removal of Short-Chain Perfluorocarboxylic Acids. *Industrial and Engineering Chemistry Research* **58**: 3329–3338. DOI:10.1021/acs.iecr.8b05506.
- Soriano, Á, D. Gorri, and A. Urtiaga. 2017. Efficient treatment of perfluorohexanoic acid by nanofiltration followed by electrochemical degradation of the NF concentrate. *Water research* **112**: 147–156. DOI:10.1016/j.watres.2017.01.043.
- Steinle-Darling, E., and M. Reinhard. 2008. Nanofiltration for trace organic contaminant removal: Structure, solution, and membrane fouling effects on the rejection of perfluorochemicals. *Environmental Science and Technology* **42**: 5292–5297. DOI:10.1021/es703207s.
- Stockholm Convention. 2017. Guidance for the inventory of perfluorooctane sulfonic acid (PFOS) and related chemicals listed under the Stockholm convention on persistent organic pollutants. :.
- Tang, C. Y., Q. S. Fu, C. S. Criddle, and J. O. Leckie. 2007. Effect of Flux (Transmembrane Pressure) and Membrane Properties on Fouling and Rejection of Reverse Osmosis and Nanofiltration Membranes Treating Perfluorooctane Sulfonate Containing Wastewater. *Environmental science & technology* **41**: 2008–2014. DOI:10.1021/es062052f.
- Tang, C. Y., Q. S. Fu, A. P. Robertson, C. S. Criddle, and J. O. Leckie. 2006. Use of Reverse Osmosis Membranes to Remove Perfluorooctane Sulfonate (PFOS) from Semiconductor Wastewater. *Environmental science & technology* **40**: 7343–7349. DOI:10.1021/es060831q.
- Thompson, S., A. M. Gutierrez, J. Bukowski, and D. Bhattacharyya. 2024. Microfiltration Membrane Pore Functionalization with Primary and Quaternary Amines for PFAS Remediation: Capture, Regeneration, and Reuse. *Molecules* **29**. DOI:10.3390/molecules29174229.
- Tian, L., P. Zhou, Z. Su, T. Liu, N. Graham, T. Bond, and W. Yu. 2023. Insights into the properties of surface waters and their associated nanofiltration membrane fouling: The importance of biopolymers and high molecular weight humics. *Chemical Engineering Journal* **451**: 138682. DOI:10.1016/j.cej.2022.138682.
- Tsai, Y. -, A. Yu-Chen Lin, Y. -. Weng, and K. -. Li. 2010. Treatment of perfluorinated chemicals by electro-microfiltration. *Environmental Science and Technology* **44**: 7914–7920. DOI:10.1021/es101964y.
- Tsai, Y., A. Yu-Chen Lin, Y. Weng, and K. Li. 2010. Treatment of Perfluorinated Chemicals by Electro-Microfiltration. *Environmental science & technology* **44**: 7914–7920. DOI:10.1021/es101964y.
- Turk, O. K., M. Cakmakci, I. H. Zengin, D. Karadag, and E. Yuksel. 2025. Improving PFAS Rejection by Ultrafiltration Membranes via Polyelectrolyte Multilayer Coating. *Membranes* **15**: 172. doi: 10.3390/membranes15060172. DOI:10.3390/membranes15060172.
- U.S. Environmental Protection Agency. 2024. EPA’s PFAS Strategic Roadmap. .
- U.S. Environmental Protection Agency. 2023. CompTox Chemicals Dashboard. <https://comptox.epa.gov/dashboard/chemical-lists/PFASSTRUCT>.

- Urbanas, D., and E. Baltrėnaitė-Gedienė. 2025. A critical review of the methods being proposed to solve the PFAS problem in drinking water: Are they practically applicable in real world? *Emerging Contaminants* **11**: 100563. DOI:10.1016/j.emcon.2025.100563.
- Van Der Bruggen, B., C. Vandecasteele, T. Van Gestel, W. Doyen, and R. Leysen. 2003. A review of pressure-driven membrane processes in wastewater treatment and drinking water production. *Environmental Progress* **22**: 46–56. DOI:10.1002/ep.670220116.
- Venkatesh Reddy, C., R. Kumar, P. Chakraborty, B. Karmakar, S. Pottipati, A. Kundu, and B. Jeon. 2024. A critical science mapping approach on removal mechanism and pathways of per- and poly-fluoroalkyl substances (PFAS) in water and wastewater: A comprehensive review. *Chemical Engineering Journal* **492**: 152272. DOI:10.1016/j.cej.2024.152272.
- Wang, T., C. Zhao, P. Li, Y. Li, and J. Wang. 2015. Fabrication of novel poly(m-phenylene isophthalamide) hollow fiber nanofiltration membrane for effective removal of trace amount perfluorooctane sulfonate from water. *Journal of Membrane Science* **477**: 74–85. DOI:10.1016/j.memsci.2014.12.038.
- Wang, Y., and C. Tang. 2011. Protein fouling of nanofiltration, reverse osmosis, and ultrafiltration membranes—The role of hydrodynamic conditions, solution chemistry, and membrane properties. *Journal of Membrane Science* **376**: 275–282. DOI:10.1016/j.memsci.2011.04.036.
- Wang, Z., A. M. Buser, I. T. Cousins, S. Demattio, W. Drost, O. Johansson, K. Ohno, G. Patlewicz, A. M. Richard, G. W. Walker, G. S. White, and E. Leinälä. 2021. A New OECD Definition for Per- and Polyfluoroalkyl Substances. *Environmental science & technology* **55**: 15575–15578. DOI:10.1021/acs.est.1c06896.
- Wu, A. X., S. Pfremmer, T. Smrz, and S. M. Smith. 2025. Benchmarking of Commercial Reverse Osmosis and Nanofiltration Membranes for Ultrashort-, Short-, and Long-Chain Perfluoroalkyl and Polyfluoroalkyl Substance Rejection. *Water Environment Research* **97**: e70227. DOI:10.1002/wer.70227.
- Wu, Y. 2023. Efficient removal of perfluorinated compounds with the polyamide nanofiltration membrane and membrane fouling resistance analysis. *Water Quality Research Journal* **58**: 215–229. DOI:10.2166/wqrj.2023.011.
- Ying Shi, C., G. Chen, and L. F. Dumée. 2024. Perfluoroalkyl substances concentration from groundwater via direct contact membrane distillation. *npj Clean Water* **7**. DOI:10.1038/s41545-024-00414-6.
- Yogarathinam, L. T., U. Baig, A. Waheed, and I. H. Aljundi. 2025. Recent progress and challenges in nanocomposite membranes for efficient removal of hazardous per-/polyfluoroalkyl substances from water sources. *Journal of Environmental Chemical Engineering* **13**. DOI:10.1016/j.jece.2025.115470.
- Yu, Y., C. Zhao, L. Yu, P. Li, T. Wang, and Y. Xu. 2016. Removal of perfluorooctane sulfonates from water by a hybrid coagulation–nanofiltration process. *Chemical Engineering Journal* **289**: 7–16. DOI:10.1016/j.cej.2015.12.048.

- Zahmatkesh, S., Z. Chen, N. A. Khan, and B. Ni. 2024. Removing polyfluoroalkyl substances (PFAS) from wastewater with mixed matrix membranes. *Science of The Total Environment* **912**: 168881. DOI:10.1016/j.scitotenv.2023.168881.
- Zeng, C., S. Tanaka, Y. Suzuki, and S. Fujii. 2017. Impact of feed water pH and membrane material on nanofiltration of perfluorohexanoic acid in aqueous solution. *Chemosphere* **183**: 599–604. DOI:10.1016/j.chemosphere.2017.05.132.
- Zhang, K., P. Cheng, Y. Liu, and S. Xia. 2024. Efficient removal of per- and polyfluoroalkyl substances by a metal-organic framework membrane with high selectivity and stability. *Water research* **265**: 122276. DOI:10.1016/j.watres.2024.122276.
- Zhang, Y., A. Thomas, O. Apul, and A. K. Venkatesan. 2023. Coexisting ions and long-chain per- and polyfluoroalkyl substances (PFAS) inhibit the adsorption of short-chain PFAS by granular activated carbon. *Journal of hazardous materials* **460**: 132378. DOI:10.1016/j.jhazmat.2023.132378.
- Zhang, Z., Y. Liu, Y. Chen, H. Qu, J. Ma, M. Pang, B. Wang, and G. Yu. 2026. Per- and polyfluoroalkyl substances in the environment and human tissue: A critical review of distribution, exposure pathways, and health risks. *Environmental Pollution* **397**: 127988. DOI:10.1016/j.envpol.2026.127988.
- Zhao, D., and S. Yu. 2015. A review of recent advance in fouling mitigation of NF/RO membranes in water treatment: pretreatment, membrane modification, and chemical cleaning. *Desalination and Water Treatment* **55**: 870–891. DOI:10.1080/19443994.2014.928804.
- Zhao, P., X. Xia, J. Dong, N. Xia, X. Jiang, Y. Li, and Y. Zhu. 2016. Short- and long-chain perfluoroalkyl substances in the water, suspended particulate matter, and surface sediment of a turbid river. *Science of The Total Environment* **568**: 57–65. DOI:10.1016/j.scitotenv.2016.05.221.
- Zhi, Y., X. Zhao, A. Shuai, Y. Jia, X. Cheng, S. Lin, F. Xiao, L. Han, H. Chai, Q. He, and C. Liu. 2025. Enhancing rejection of short-chain per- and polyfluoroalkyl substances by tailoring the surface charge of nanofiltration membranes. *Water research* **272**. DOI:10.1016/j.watres.2024.122931.
- Zhi, Y., X. Zhao, A. Shuai, Y. Jia, X. Cheng, S. Lin, F. Xiao, L. Han, H. Chai, Q. He, and C. Liu. 2025. Enhancing rejection of short-chain per- and polyfluoroalkyl substances by tailoring the surface charge of nanofiltration membranes. *Water research* **272**: 122931. DOI:10.1016/j.watres.2024.122931.
- Zhou, W., and L. Song. 2005. Experimental Study of Water and Salt Fluxes through Reverse Osmosis Membranes. *Environmental science & technology* **39**: 3382–3387. DOI:10.1021/es0403561.